Procedural Knowledge Layer

Write

Human & agentic coding

editors, with the aip extension

VS Code · Cursor · JetBrains

```
Route by_tone "Who replies" {
  Read triage_decision.tone
  Confidence_Below 0.9 -> escalate
  angry -> escalate
  calm  -> reply
}
```

syntax colours, the compiler's squiggles, a live graph preview

agents with the aip skill

Claude Code · Codex · ...

Autonomous agent

built-in procedures that write one

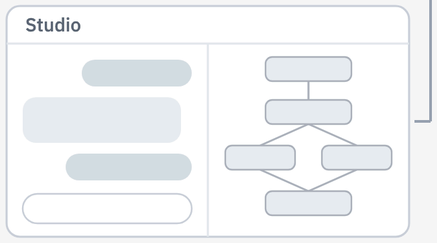
```
aip run make_procedure
-i {name, description, materials}
```

also build_procedure · edit_procedure

No-code

AIP console

the Studio: a chat beside its step graph



proc/

procedure.aip
scripts/
assets/
references/

Compile

`aip compile proc/`

lossless: aip export gives proc/ back

proc.rl

source files graph

id 6e98…aad2

publish

opened in process

Run

Claude Code · Codex · ADK · LangGraph · Strands · MAF · ...

CLI + skill · MCP · SDK

`aip runah`

the built-in harness: a chat model takes the agent's turns

`aip run`

run record `.json`

record

pull

at each step, on this machine

`Execute`	its script runs next to your data
`Decide`	asks a decision model
`Task_Agent`	pauses; the agent answers with resume

every stop writes the run file on this machine; with a server, it is posted there too

Store

Local filesystem
always

`.rl` `.json`

files under ~/.aip

alone: runs by path, no catalog, no audit

Local server
starts on demand



procedures and runs in one graph

or

Remote server
shared, with a token



procedures and runs in one graph

relate or neo4j / Aura

embedded · over Bolt

catalog

`publish` · `search` · `revisions & pins` · `pull / get`

Audit

`aip audit`

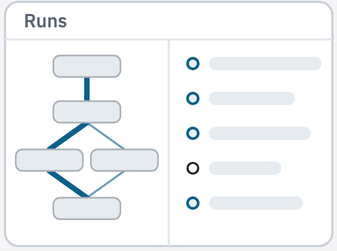
named queries, or your own Cypher

`aip export runs`

one .rl file, queried with no server

AIP console

step graph with run traffic, history, audit



A procedure is a small program a compiler can check

Seven kinds of step, two kinds of loop, no expressions. The procedure is a finite state machine; what makes it useful is what its steps call out to: a decision model, a script, an agent.

Seven kinds of step

`Start` `Decide` `Route` `Aggregate` `Execute` `Task_Agent` `End`

Every step is a card: what it reads, its own lines, what it produces, where it goes.

No expressions, by design

No arithmetic, no logical operators, no function calls; text is built only by `{key}` interpolation. Computation is an `Execute` step.

Every run ends

Every cycle is a loop with a cap (`At_Most n`), so a run ends in a number of steps you can compute from the file. Not Turing-complete, on purpose.

Keys are written once

Along any path a key has one producer. Two exceptions: the arms of a route may each write it, and a loop body may rewrite a key from before the loop (a fold). A `For` keeps per-pass keys only with `Collect … Into`.

Conditions are on answers

A `Route`, an `Until` or an aggregation reads one answer; nothing else in the state can be branched on. A decision model's answers carry a confidence the procedure can check.

Logic lives in scripts

Anything that computes is a script in `scripts/`: JSON in on stdin, JSON out on stdout. Python by default; any interpreter with a `Command` line.

One procedure, every kind of step

From `examples/language/ticket-triage`: a batch of tickets, each classified by a model, routed, answered by the agent or escalated by a script.

```
For each_ticket Over tickets As ticket, At_Most 50, Collect reply Into replies {
  Decide classify classify_question {
    Read ticket
    Answer_Questions {
      billing: true_or_false "Is this a billing question?"
      tone:    choice [calm, angry] "How does the customer sound?"
    }
    Produce classification
    Next_Step by_tone
  }

  Route by_tone "Calm customers get a drafted reply; angry or uncertain ones are escalated." {
    Read classification.tone
    Confidence_Below 0.6 -> escalate
    calm  -> draft
    angry -> escalate
  }

  Task_Agent draft "Write a reply that follows the policy." {
    Read ticket, policy
    Produce reply: string
    Next_Step revise
  }

  Execute escalate "Open a ticket in the tier-2 queue." {
    Read ticket
    Run_Script scripts/escalate.py
    Produce reply: string
            queue: string
    Continue
  }
```

STEP	WHO DOES THE WORK
`Start`	the caller: typed inputs
`Decide`	a decision model answers the questions, with confidence
`Route`	the runtime branches on one answer; `Confidence_Below` sends unsure ones elsewhere
`Aggregate`	the runtime combines answers: `All`, `Any`, `None`
`Execute`	a script, JSON in and JSON out
`Task_Agent`	the agent; the only step that pauses the run
`End`	declares the result keys

LOOP	WHAT IT IS
`For x Over list`	independent passes, one per item; batched
`Loop x Until a`	passes that build on each other until an answer holds
`Loop x n Times`	exactly n passes

Every loop has a cap; at the cap it fails the run or exits, as written.

Answers carry their confidence

A decision model answers with a value and how sure it is. The procedure can check that number; the compiler makes sure it only checks numbers that exist.

```
# what a Decide writes, one answer per question
classification.billing  {value: true,    confidence: 0.82, noul: 0.91}
classification.tone     {value: "angry", confidence: 0.9,
                         probabilities: {angry: 0.9, calm: 0.1}}
```

A ROUTE CHECKS IT

```
Route by_tone "Who replies" {
  Read triage_decision.tone
  Confidence_Below 0.9 -> escalate
  angry -> escalate
  calm  -> reply
}
```

Below 0.9, or with no confidence at all, the route goes to `escalate`.

AN AGGREGATION SETS A BAR ON EACH ANSWER

```
Aggregate summarize_check "Every policy holds, and the model was sure of each." {
  Read verdicts
  Aggregations {
    all_hold: All verdicts.follows With Confidence_Above 0.7
  }
  Produce check_summary
  Next_Step fix
```

The result is a `true_or_false` with no confidence, on purpose: the language never makes one confidence out of several.

WHO ANSWERED DECIDES WHAT CAN BE CHECKED

SOURCE	STEP	CONFIDENCE CHECK
measured	`Decide`	yes
scripted	`Execute`	yes, if the script gave one
agent	`Task_Agent`	no: `confidence_on_agent_answer`
input · aggregated	`Start` · `Aggregate`	no: a compile error

A procedure is a folder

The same shape as an Agent Skill: the procedure beside the scripts, assets and references it uses. The compiler checks that every path it names is there.

```
ticket-triage/
├── procedure.aip      the steps
├── scripts/           code an Execute step runs
│   ├── escalate.py
│   └── policy_check.py
├── assets/            prompts and files a step uses
└── references/        documents a task may load
```

Limits today

- One `Requires` per procedure, checked before the first script runs
- A script other than Python needs a `Command` line
- An asset reaches a step as a file, not a key: data the state needs comes in at `Start` or from a script that reads it
- A script runs as you, with your credentials: read it before you run it

A SCRIPT'S CONTRACT

```
# stdin: one JSON object
{"state":   {"reply": "…", "policy": "…"},   # the keys it reads
 "assets":  {"fix": "…"},                    # the files it uses
 "expects": {"passed": "true_or_false"}}     # what it must produce

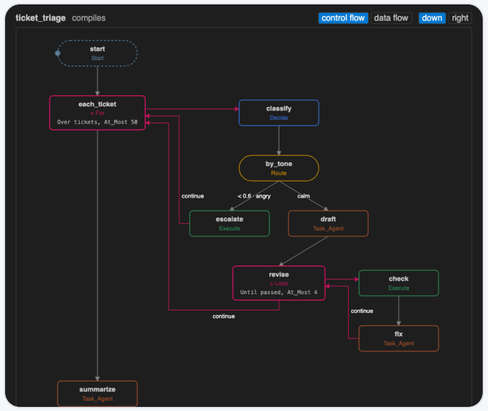
# stdout: exactly those keys
{"passed": true}
```

- Run as `python scripts/<file>`, or by its `Command`
- Files go under `$AIP_WORKDIR`, the run's working directory
- 300 seconds per script by default (`aip config --script-timeout`)
- A non-zero exit or bad JSON fails the step; `aip resume` retries it

Four ways to write one, one file at the end

Whoever writes it, the result is the same folder, checked by the same compiler (`aip compile --check`).

Editors



- VS Code extension: colours, the compiler's squiggles, a live graph preview
- Cursor and Windsurf take the same .vsix
- JetBrains and Sublime: colours, from the TextMate grammar

Coding agents

```
$ aip compile --check proc/
procedure.aip:30:3: error
  missing_branch: Route by_tone
  has no branch for angry
```

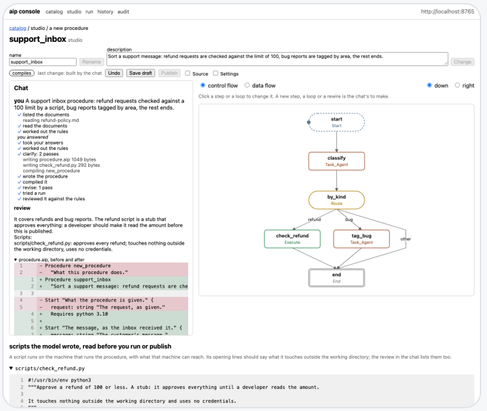
- Claude Code, Codex, any agent with the aip skill
- The skill carries the language on one page (`references/language.md`)
- The agent writes, compiles, fixes until it compiles

Autonomous agent

```
aip run make_procedure
-i {name, description,
    materials}
```

- A built-in procedure that writes a procedure: the agent writes, the compiler checks, round until clean
- `build_procedure` reads documents and asks a person the open questions; `edit_procedure` changes one

No-code: the Studio



- In the AIP console: a chat that runs build and edit, the step graph beside it
- Forms for field edits; the scripts a model wrote are shown before anything runs

COMPILE

aip compile: check everything, then write one file

Nothing runs unchecked: running a folder compiles it first, and a bundle is verified again every time it is opened.

parse	check	collect	identify	graph	write
the syntax tree	44 diagnostic codes	scripts, assets, references	sha256 of source + files	steps, flow, data	proc.rl

ONE MISTAKE, ONE ERROR, AT ITS LINE

```
$ aip compile --check ticket-triage/
procedure.aip:30:3: error missing_branch: Route by_tone has no branch
  for angry; a route names every value classification.tone can take
procedure.aip:42:3: error missing_requires: Execute escalate runs a
  script, but the header declares no Requires
procedure.aip:77:26: error unavailable_key: queue is produced inside
  For each_ticket, whose keys are per-pass and gone when the loop exits;
  Collect queue Into <list> keeps them
procedure.aip: 3 errors
```

The same JSON feeds the editor's squiggles and the Studio's graph.

What the checker catches

`unreachable_step`	nothing leads to a step
`cycle_without_loop`	a cycle outside a loop
`missing_cap`	a loop with no At_Most
`unavailable_key`	a key some path never produced
`duplicate_producer`	a key written twice on one path
`missing_branch`	a route misses a value
`confidence_on_agent_answer`	checking a number that isn't there
`missing_file`	a script or prompt that isn't in the folder

The bundle

- One file: the source byte for byte, every file, and the graph
- Its id is the hash of the source and the files, nothing else: same folder, same id
- Lossless: `aip export` writes the folder back

COMPILE · RELATE

Relate: a graph database in one file

The bundle, the local store and an exported audit are each a Relate file: a whole Cypher graph you can copy, commit or mail, opened in process with no server.

relate — Neo4j Labs' embedded graph engine

Cypher, embedded

- An openCypher engine in a single binary, written in Rust
- 100% of the openCypher TCK 2024.3 (3,892 of 3,892 scenarios), Cypher 25 features
- In process from Python, JavaScript (Node and WASM), Java, Go and Rust

One file, or a server when you want one

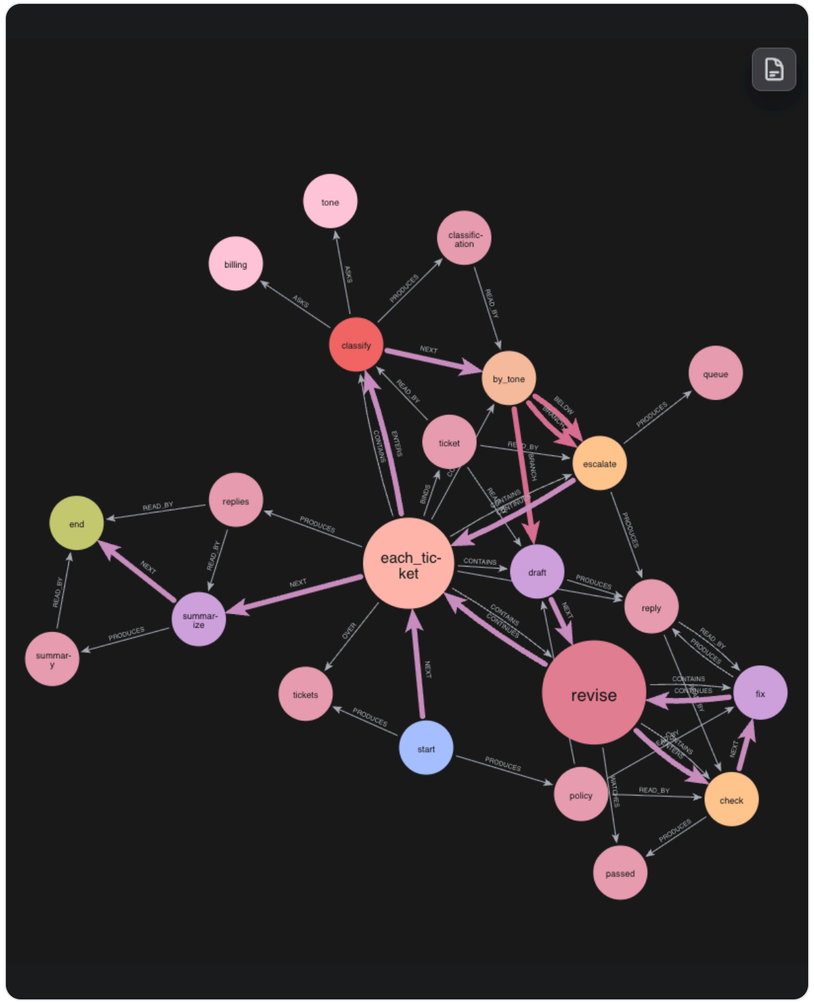
- A store is a directory or one file; durable with a write-ahead log
- Serves the HTTP Query API, MCP and Bolt on demand, so Neo4j Browser connects
- Pulls a subgraph from Neo4j or Aura, and pushes one back

How AIP uses it

- The compiler writes the bundle's graph with one body of Cypher
- The runtime loads the bundle into memory and reads each step with a Cypher query as the run reaches it
- The server's store is a Relate file by default; the same statements run on Neo4j or Aura

The bundle is a graph: steps, control flow, data flow

`ticket-triage.rl` in Neo4j Browser, served read-only by Relate over Bolt. The same graph the console draws.



GROUP	NODES AND EDGES
Nodes	`Procedure` `Step` (+ its kind) `Loop` `Data` `Question` `Aggregation` `File`
Control flow	`NEXT` `BRANCH {value}` `BELOW {below}` `ENTERS` `CONTINUES` `CONTAINS`
Data flow	`PRODUCES` `READ_BY {field, source}`
Loops	`OVER` `BINDS` `COLLECTS` `WATCHES`
Decisions	`ASKS` `AGGREGATES`
Files	`RUNS` `PROMPTED_BY` `USES` `HAS_FILE`

Every node carries the bundle id, so many bundles share one store.

EVERY EDGE A RUN CAN TAKE

```
MATCH (p:Procedure {name: 'ticket_triage'})
MATCH flow = (p)-[:STARTS_AT]->()
  -[:NEXT|BRANCH|BELOW|ENTERS|CONTINUES*]->()
RETURN flow
```

WHO PRODUCES WHAT, AND WHO READS IT

```
MATCH (w)-[:PRODUCES]->(d:Data)-[e:READ_BY]->(r)
RETURN w.name AS producer,
       d.name AS data, r.name AS reader
```

WHICH ROUTES AN AGENT STEERS

```
MATCH (r:Route)<-[e:READ_BY]-(d:Data)
WHERE e.source = 'agent'
RETURN r.name, d.name
```

RUN

aip run walks the graph, one step at a time

A run loads the bundle into memory and follows it on the machine that runs it: scripts next to the data, decisions from a model, tasks handed to the agent. It stops when it pauses, ends or fails.

STEP	AT RUN TIME
`Start`	checks the inputs against their declared types
`Decide`	sends the questions to the decision model; records every answer whole. No model, no answer: it never answers by hand
`Route`	fires its confidence check when the answer is below the bar or has none; otherwise branches on the value
`Aggregate`	applies All, Any or None over the answers, with an optional bar on each
`Execute`	checks Requires, runs the script from the cache, reads its JSON
`Task_Agent`	renders the prompt and pauses for the agent
`End`	returns the keys the path produced

Loops

- A cap is checked before the first pass; at the cap the run fails or exits, as written
- `For` passes run in batches of 10 (`--batch`)
- A batch whose body is one `Decide` is **one** call to the model
- Passes waiting at one step go out as one pause with several tasks
- Batching never changes a result

Stays put

- A run stays on the exact bundle it started on
- Every stop writes the run file; with a server, it is posted there too

The agent's turn: pause, answer, resume

Only a `Task_Agent` pauses. The run hands over the task and waits; the answer is checked strictly before anything moves.

WHAT THE AGENT RECEIVES

```
{"paused": "task", "step": "triage",
 "tasks": [{
   "index": 0,
   "prompt": "Read the customer's message and classify it. …",
   "state": {"message": "I was charged twice for March."},
   "references": [],
   "answer_with": {"billing": "true_or_false", "tone": "choice"},
   "labels": {"tone": ["angry", "calm"]}}],
 "hint": "answer with {\"billing\": true | false, …}",
 "run_id": "4f1c9a2e7b3d5068"}
```

WHAT IT SENDS BACK

```
$ aip resume 4f1c9a2e7b3d5068 --answers '{"billing": true, "tone": "calm"}'
```

Strict

- A missing key, an extra one, or a value outside the labels is `invalid_output`: nothing is applied, the pause stands
- Several tasks at once (a batch) are answered together, keyed by `index`
- Inside a capped loop a task says `pass_of` and `last_pass`

Errors say what to do

`script_failed` `timeout` `requires_unmet`	fix the cause, resume: the step runs again
`no_decision_model` `decision_model_failed`	configure the model, resume
`invalid_output`	answer again
`loop_cap`	the run is over

RUN · ANY AGENT

Any agent can drive a run, with no new code

Claude Code, Codex, ADK, LangGraph, Strands, Microsoft Agent Framework: anything with a shell or MCP. The procedure decides what happens; the agent answers its tasks.

CLI + the aip skill

```
aip search "charged twice"
aip info billing_support
aip run billing_support -i …
aip resume <run> --answers …
```

- `aip skill install` puts the skill in the agents on this machine
- The skill tells the agent: search, run, answer pauses, never do the steps by hand

MCP

```
aip mcp              # stdio
aip mcp --http PORT  # HTTP
```

- Seven tools: `search` `describe` `compile` `run` `resume` `runah` `pull`
- A pause or an error is a tool result, not an MCP error
- The client's name becomes the run's agent

Python SDK

```
result = client.run(
  "billing_support",
  {"message": "…"})
while isinstance(result,
                 aip.Paused):
  result = client.resume(
    result, answers_for(result))
print(result.result)
```

- Every command is one call: run, resume, runah, search, publish, audit…

aip runah: the built-in harness

- The runtime takes the agent's turns: each task is one short chat with a model
- Five tools in the working directory: `read_file` `list_files` `search` `write_file` `compile`
- Any Chat Completions endpoint; Anthropic's by default
- Stops at `--human` steps for a person; capped per task (40 tool calls, 500k tokens)
- Recorded as answered by runah, work `observed`; an outside agent's is `reported`

STORE · PROCEDURAL KNOWLEDGE LAYER

Files always; one server when you want more

Every run writes its files on the machine that ran it. Point the client at a server, local or remote, and procedures and runs land in one graph: a catalog to search and a record to audit.

Local filesystem · always

```
~/.aip/bundles/<id>.rl
~/.aip/runs/<run_id>.json
~/.aip/cache/<id>/
```

- Every bundle a run used, every run file, scripts unpacked once per bundle
- Alone: run by path, no catalog, no audit

Local server · starts on demand

```
aip config server local
```

- The installer's default; listens on `127.0.0.1:8247`, loopback only, no token
- Started by the first command that needs it, stopped after 15 idle minutes
- One Relate file: `~/.local/share/aip/server/store.rl`

Remote server · shared

```
aip config server URL --token T
```

- Tokens with scopes `read`, `run`, `publish`; a token's name is the run's `user`
- Relate, or Neo4j / Aura over Bolt when a team shares it
- A run never fails because a server is down: the stop is just not recorded

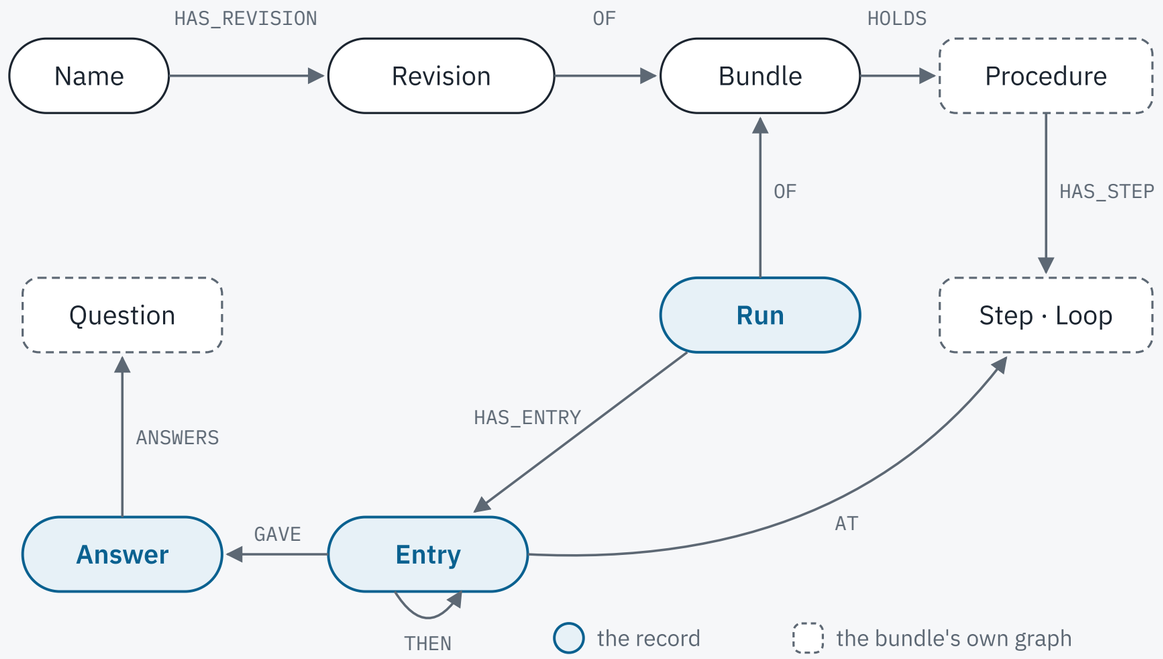
The catalog

- `aip publish` verifies the bundle (hashes, id, graph) and gives it the next revision of its name
- A bundle is stored by its id: publishing the same content twice changes nothing
- `aip search`: BM25 over names, descriptions, Use_When lines and steps; an exact name wins
- `name`, `name@3` or `name@latest`; `aip pin` and `aip retire` (retire wins over pin)
- A run of a folder is recorded as a draft unless that exact bundle is published
- Ships with `make_procedure`, `build_procedure`, `edit_procedure`

STORE · ONE GRAPH

Procedures and runs in one graph

The server stores each bundle's graph as compiled and writes the record beside it. Every history entry points at the step it ran, so a question about runs is a question about the procedure too.



What an entry keeps

- Every entry: step, kind, pass, when, how long, what it read and produced
- A `Decide`: the model, its answers with confidence and probabilities
- A `Route`: what it read, where it went, whether its check fired and on what numbers
- A `Task_Agent`: who answered it, and what answering took

Two backends, one set of statements

relate embedded, one file, one writer

neo4j / Aura over Bolt, several servers may share it

The same Cypher writes, searches and audits on both.

AUDIT

Questions over the record, by name or in Cypher

`aip audit` runs named queries over the store; the console shows the same lists. Every query is read-only: it runs in a transaction that is never committed.

Over the recent runs

`confidence-fired`	how often each route's check fired, and why
`model-answers`	what the model answered, least sure first
`failing-aggregations`	aggregations that did not hold
`failing-scripts`	scripts runs stand failed at
`agent-work`	tasks, tool calls and tokens per agent step
`untaken-branches`	branches no run took
`repeated-cycles`	loops that went round more than expected

Over each published procedure

`agent-routes`	routes an agent's answer steers
`lineage`	every key a result is derived from
`cycles`	every loop and what bounds it
`unused-data · empty-results · missing-fields`	procedures to tidy

YOUR OWN: ROUTES WHOSE CONFIDENCE CHECK FIRED

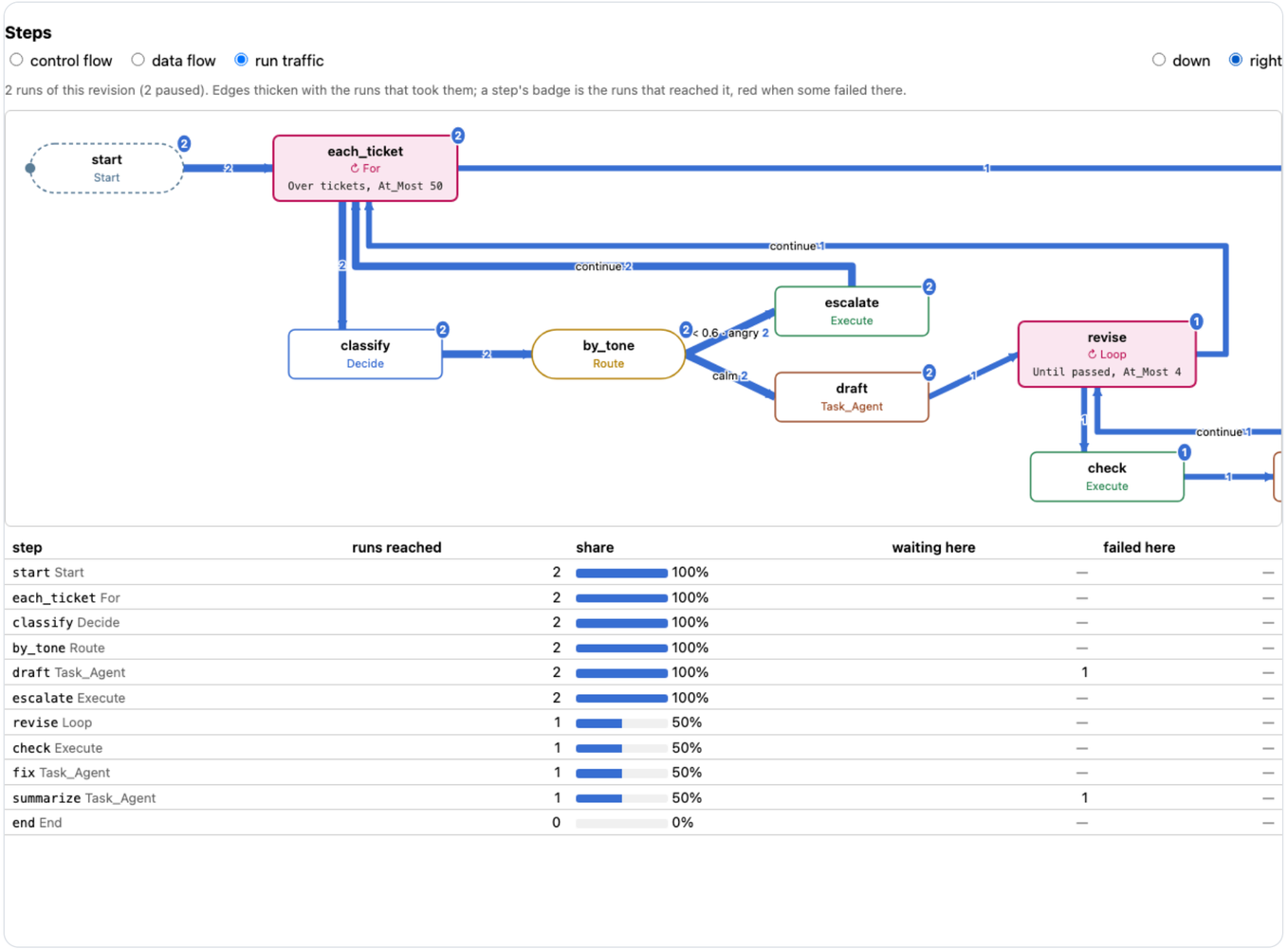
```
MATCH (r:Run)-[:HAS_ENTRY]->(e:Entry {kind: 'Route'})
      -[:AT]->(route:Route)
WHERE e.check_fired = true
RETURN r.name AS procedure, route.name AS route,
       e.check_reason AS reason,
       e.check_confidence AS confidence,
       e.check_below AS bound, e.to AS went_to
```

```
$ aip audit --save fired-checks --cypher '…'
$ aip audit fired-checks --name ticket_triage --window 100
```

A stored query is refused if it writes or does not parse (checked with EXPLAIN). Filters: `name` `agent` `session` `user` `draft` `window`.

The AIP console, and one file for an auditor

The console draws the record over the step graph; an export puts the runs and the procedures they ran in one file anyone can query with no server.



step	runs reached	share	waiting here	failed here
start Start	2	100%	—	—
each_ticket For	2	100%	—	—
classify Decide	2	100%	—	—
by_tone Route	2	100%	—	—
draft Task_Agent	2	100%	1	—
escalate Execute	2	100%	—	—
revise Loop	1	50%	—	—
check Execute	1	50%	—	—
fix Task_Agent	1	50%	—	—
summarize Task_Agent	1	50%	1	—
end End	0	0%	—	—

AIP console

- Run traffic: each edge as thick as the share of runs that took it; badges for runs reached, red where some failed
- A run's page: its path over the graph, every decision's confidence, every check that fired
- History by procedure, status, agent, session, user; the audit queries as lists

Export

```
$ aip export runs ticket_triage --since 2026-11-01 -o audit.rl
$ relate query --database audit.rl \
    "MATCH (r:Run) RETURN count(r)"
```

- The matching runs, the graphs of the bundles they ran, and their revisions
- Every query that works on the server works on the file

LINKS

Where it lives

The private repositories are current, Relate included. The public ones are behind, at 0.7, and do not mention Relate.

Private · current (0.9, with Relate)

`github.com/zach-blumenfeld/aip-dev`

language, compiler, runtime, server, console and VS Code extension, combined in one repo; branch release-0.9.0

this deck

`docs/slides/` and the figure in `docs/figures/`

Public · behind (0.7, no Relate)

`github.com/zach-blumenfeld/aip`	v0.7.0, 2 Nov 2026
`github.com/zach-blumenfeld/aip-console`	v0.7.1, 3 Nov 2026
`pypi.org/project/agent-instruction-protocol`	0.7.0
`npmjs.com/package/aip-console`	0.7.1